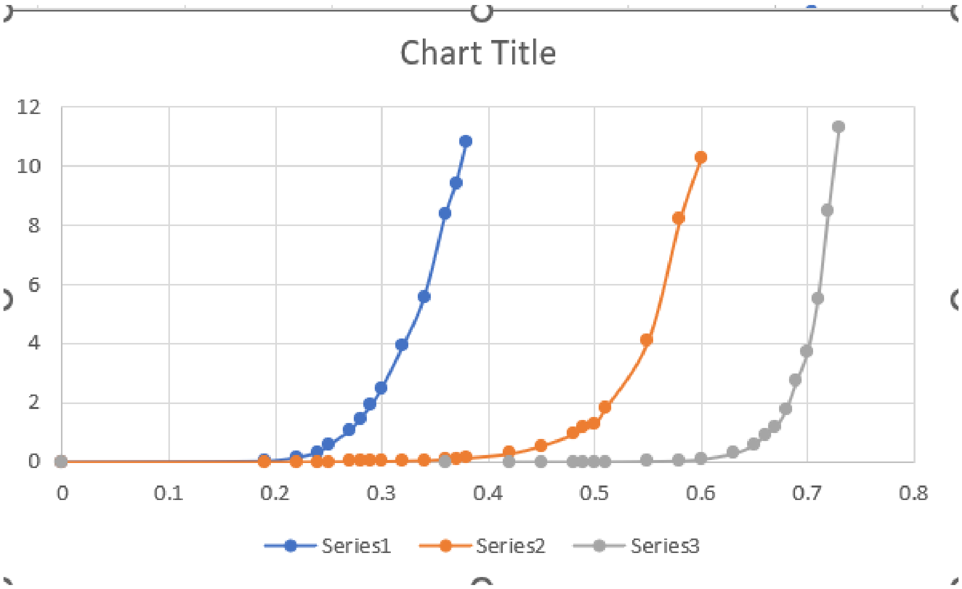
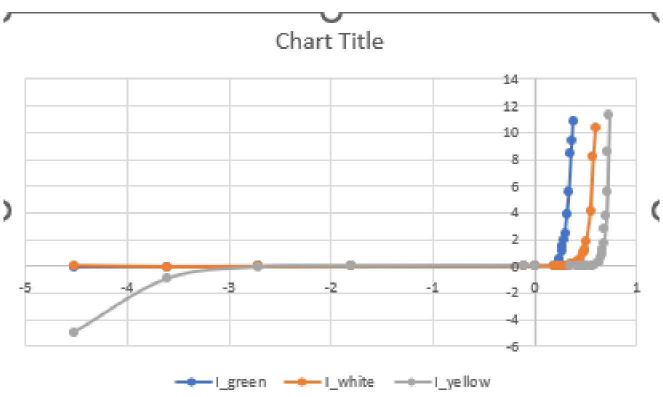
Question 1

I-V readings for the green diode:

V	I_green	I_white	I_yellow
-4.52	-0.048	0.048	-5
-3.61	-0.039	-0.039	-0.98
-2.71	-0.03	-0.03	-0.092
-1.8	-0.021	-0.02	-0.022
-0.1	-0.012	-0.003	-0.004
0	0	0	0
0.19	0.04	0.002	
0.22	0.16	0.004	
0.24	0.33	0.007	
0.25	0.56	0.01	
0.27	1.07	0.015	
0.28	1.46	0.018	
0.29	1.92	0.021	
0.3	2.48	0.024	
0.32	3.92	0.04	
0.34	5.56	0.055	
0.36	8.36	0.088	0.001
0.37	9.43	0.102	
0.38	10.85	0.135	
0.42		0.284	0.002
0.45		0.548	0.003
0.48		0.937	0.004
0.49		1.154	0.005
0.5		1.27	0.006
0.51		1.81	0.007
0.55		4.1	0.022
0.58		8.21	0.046
0.6		10.3	0.093
0.63			0.284
0.65			0.574
0.66			0.924
0.67			1.17
0.68			1.76
0.69			2.73
0.7			3.71
0.71			5.49
0.72			8.48
0.73			11.3





Here, the blue curve is for the green diode, the orange for the white diode, and the grey for the yellow diode.

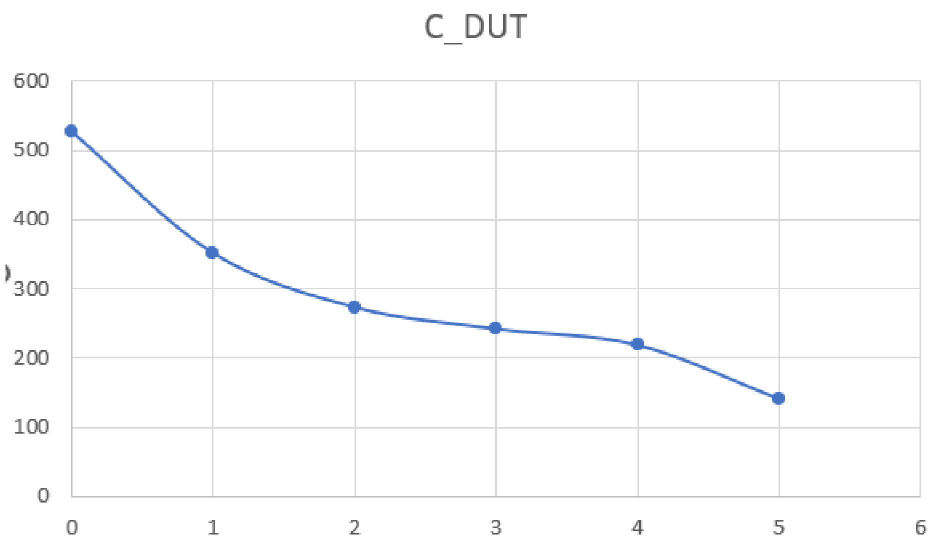
The blue curve is for a Schottky diode as evident from the very low cut-in voltage (around 0.2V)

The orange curve is for a photodiode since it's cut-in voltage is around 0.5V

The grey curve must be a PIN diode with a cut-in voltage around 0.7 V

Question 2

V_dc	C_DUT
0	526.5
1	351
2	273
3	241.8
4	218.4
5	140.4



C_DUT is in pF here